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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics	
not required.	
Inclusion and Exclusion Criteria	
not required.	
Attrition	
not required.	
Sex as a biological variable	
not required.	
Subject Demographics	
Age: not required.	
Weight: not required.	
Randomization	
Because equilibrium of the sample size between the UC group and the normal group is required for randomized study	
Blinding	
not detected.	
Power Analysis	
not detected.	
Replication	
not required.	
Data Information	
Identifiers: Data collection and processing Five UC-related gene datasets, including	GSE128682

GSE128682, GSE134025, GSE87473, GSE102746, and GSE59071, were curated from the Gene Expression Omnibus (GEO, https:www.ncbi.nlm.nih.govgeo).	GSE134025
	GSE87473
	GSE102746
	GSE59071

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and Algorithms			
Data collection and processing Five UC-related gene datasets, including GSE128682, GSE134025, GSE87473, GSE102746, and GSE59071, were curated from the Gene Expression Omnibus (GEO, https:www.ncbi.nlm.nih.govgeo).	Gene Expression Omnibus		Suggestion: Gene Expression Omnibus (GEO) (RRID:SCR_005012)(link)
The effect between any two batches was eliminated by ComBat using the Surrogate Variable Analysis (SVA) package in R.	ComBat		Suggestion: ComBat (RRID:SCR_010974)(link)
DEG screening Differentially expressed genes (DEGs) between UC samples and normal samples in the training cohort were screened using the limma package.	limma		
The ggplot2 package and the pheatmap package were used to perform the visualization of the DEGs and to draw the heatmap , respectively . 2.3 .	pheatmap		Suggestion: pheatmap (RRID:SCR_016418)(link)
KEGG) pathway enrichment analysis were accomplished by the clusterProfiler package.	KEGG		
The visualization of enrichment results was achieved using the ggplot2 package in R and Cytoscape Version 3.7.1 . 2.4 .	ggplot2		Suggestion: ggplot2 (RRID:SCR_014601)(link)
	Cytoscape		
PPI network construction By utilizing the Search Tool for the Retrieval of Interacting Genes (STRING) , the protein-protein interaction network (PPI) was established to further investigate the function of DEGs at the protein level .	STRING		Suggestion: STRING (RRID:SCR_005223)(link)
GSEA enrichment analysis Gene Set Enrichment Analysis (GSEA) was performed on the gene expression matrix through clusterProfiler package to identify the immune-related pathways 21 .	clusterProfiler		Suggestion: clusterProfiler (RRID:SCR_016884)(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
Wilcoxon rank-sum test was used to assess the differences in the proportion of IICs between UC and control.	Wilcoxon rank-sum test

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For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq